

EcoOrbit: A Hybrid Physics–AI Decision Engine for Predictive De-Orbit Optimization of LEO Small Satellites

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Abstract—The unmanaged proliferation of orbital debris in Low Earth Orbit (LEO) imposes a growing operational, economic, and environmental burden on the global satellite industry, with more than 36,500 tracked debris objects, over one million untracked sub-decimeter fragments, and a collision-avoidance and insurance burden exceeding USD 50 million per year. End-of-life satellite disposal is still treated largely as a reactive operation, governed by static guidelines rather than predictive analytics. This paper introduces *EcoOrbit*, a hybrid physics–AI decision engine that converts de-orbit timing into a predictive, multi-objective optimization problem. A MATLAB numerical propagator integrates the equations of orbital decay under quadratic atmospheric drag using a one-hour Euler step. A degree-three polynomial regression, trained on the propagator output in Python, predicts Remaining Useful Life (RUL) in microseconds, replacing a four-second numerical simulation while retaining a coefficient of determination $R^2 = 0.9828$ and a root-mean-square error of 8.58 days. A weighted cost function balancing fuel expenditure, residual collision risk, and post-mission debris dwell time identifies the optimal de-orbit day. Validation against the King-Hele closed-form analytical solution shows agreement within 3.2 percent across all three estimators. For the reference 250 kg LEO small satellite, *EcoOrbit* recommends de-orbit at day 200, removing 62 percent of post-mission debris dwell time. The architecture is fleet-scalable: 10^4 satellites can be re-evaluated in under one second on commodity hardware, supporting future global LEO traffic-management services.

Index Terms—Atmospheric drag, debris mitigation, de-orbit, King-Hele equation, Low Earth Orbit (LEO), multi-objective optimization, orbital decay, polynomial regression, predictive maintenance, Remaining Useful Life (RUL), space situational awareness, sustainability.

NOMENCLATURE

A	Reference cross-sectional area, m^2 .
a	Orbital semi-major axis, m .
C_D	Drag coefficient, dimensionless.
$d(t)$	Normalized post-mission debris dwell cost.
$f(t)$	Normalized fuel/ ΔV cost.
F_D	Atmospheric drag force, N .
H	Atmospheric scale height, m .
h	Geometric altitude above mean sea level, m .
$J(t)$	Multi-objective cost function, dimensionless.
m	Spacecraft mass, kg .
R_E	Earth equatorial radius, m .
$r(t)$	Normalized residual collision risk.
t^*	Optimal de-orbit day.
t_{RUL}	Remaining useful life, days.
v	Orbital velocity, m/s .
β	Ballistic coefficient, $\text{m}/(C_D A)$, kg/m^2 .
μ	Earth gravitational parameter, m^3/s^2 .
ρ	Atmospheric density, kg/m^3 .
w_f, w_r, w_d	Cost-function weights.

I. INTRODUCTION

Low Earth Orbit (LEO) has become the most economically critical orbital regime of the twenty-first century. It hosts the International Space Station, the majority of Earth-observation assets that monitor climate change, and the growing mega-constellations that deliver global broadband connectivity. The same accessibility that makes LEO commercially attractive, however, has produced an environment in which thousands of intact satellites and over one million catalogued and

uncatalogued fragments share an increasingly congested set of altitude shells [1], [2].

Surveillance networks tracked more than 36,500 objects larger than 10 cm in 2024, while statistical models estimate the population of lethal but invisible 1–10 cm fragments at well above one million [3]. At a typical LEO closing velocity of approximately 7.8 km s^{-1} , even a 1 cm aluminum fragment carries the kinetic energy of a hand grenade. Collision-avoidance maneuvers, hardened spacecraft design, and on-orbit insurance now impose more than USD 50 million per year in incremental cost on operators of commercial constellations [4].

The long-term failure mode of an unmanaged LEO is the Kessler cascade [5]: a hyper-velocity collision creates fragments, each of which becomes a future projectile, until debris density exceeds a self-sustaining threshold and the affected shell becomes unusable for centuries. Mitigation guidance, codified by the Inter-Agency Space Debris Coordination Committee (IADC) [6], typically recommends post-mission disposal within 25 years, but enforcement is uneven and the underlying timing decision is rarely optimized.

This paper proposes *EcoOrbit*, a predictive de-orbit decision engine that reframes end-of-life disposal as a quantitative,

multi-objective optimization problem. The contributions are fourfold:

1) a hybrid architecture that fuses a MATLAB numerical orbital-decay simulator with a Python machine-learning regressor, and connects them to a transparent multi-objective cost function;

2) a Remaining Useful Life (RUL) predictor that achieves $R^2 = 0.9828$ and 8.58-day RMSE while delivering a more than 10^6 inference speed-up over the underlying numerical propagator;

3) a triangulated validation strategy that cross-checks the numerical propagator against the King-Hele analytical lifetime [7] and the AI regressor, with an inter-method spread of 3.2 percent;

4) a fleet-scale deployment analysis showing that 10^4 satellites can be evaluated in under one second on commodity hardware, enabling near-term constellation-wide adoption.

The remainder of the paper is organized as follows. Section II reviews the orbital debris environment and the modeling traditions on which EcoOrbit builds. Section III describes the system architecture and data flow. Section IV develops the mathematical foundation. Sections V–VII present the MATLAB numerical layer, the Python intelligence layer, and the multi-objective optimizer, respectively. Section VIII reports results and triangulated validation. Sections IX–XI discuss limitations, future work, and sustainability impact. Section XII concludes.

II. BACKGROUND AND RELATED WORK

A. The LEO Debris Environment

The orbital debris population is governed by a competition between two source-and-sink processes. Sources include intact spacecraft and rocket bodies left in orbit, fragmentation from explosions and collisions, and surface-degradation products such as paint flakes and multi-layer-insulation debris [1]. The dominant sink in LEO is atmospheric drag: residual neutral density at 200–800 km altitudes provides a small but persistent decelerating force that ultimately leads to re-entry. The lifetime of a given object depends sensitively on its ballistic coefficient $\beta = m / (Cd A)$, on solar activity through its modulation of upper-atmosphere temperature, and on initial orbit altitude [2], [8].

B. Existing De-Orbit Strategies

Operational disposal strategies fall into three broad categories [9]: (i) controlled re-entry using residual propellant, applied to large platforms with targeted debris-footprint constraints; (ii) altitude reduction to a graveyard or naturally decaying orbit, common for small satellites without on-board propulsion; and (iii) passive devices that increase the effective drag area, including drag sails, electrodynamic tethers, and deployable balloons [10], [11]. Each strategy implies a timing decision that has historically been governed by mission planning and regulatory deadlines rather than by an optimization that explicitly trades fuel cost, residual collision risk, and post-mission dwell time.

C. Predictive Modeling Approaches

Closed-form solutions in the King-Hele tradition [7] approximate orbital lifetime under exponential atmospheric models with sufficient accuracy for first-cut design but neglect higher-order perturbations. High-fidelity propagators such as the SGP4/SDP4 family, NASA's High Accuracy Satellite Drag Model, and ESA's MASTER [3] capture additional physics at substantially higher computational cost. More recently, machine-learning surrogates trained on simulator output have been proposed to accelerate constellation-scale debris evolution [12], [13]; however, surrogates are usually presented either in isolation or as black-box accelerators. EcoOrbit differs in that it (a) couples the surrogate to a transparent analytical baseline for cross-validation and (b) embeds the surrogate inside a multi-objective decision rule rather than using it only as an end prediction.

D. Regulatory Context

Internationally, the IADC's 25-year post-mission disposal guideline [6] has been widely adopted but is not strictly enforced. The U.S. Federal Communications Commission's recent five-year rule [20] for U.S.-licensed LEO satellites tightens compliance considerably and signals a global trend toward shorter dwell limits. The United Nations Long-Term Sustainability of Outer Space Activities Guidelines [22] formalize a complementary set of voluntary best practices that address coordination, transparency, and capacity-building. EcoOrbit is designed to operate within and beyond all three frameworks simultaneously: its decision logic is exposed and auditable, and its multi-objective formulation naturally accepts disposal-deadline constraints as additional terms in $J(t)$.

E. Constellation-Scale Risk

Recent collision-probability analyses for proposed broadband constellations [21] indicate that pairwise conjunction rates scale super-linearly with constellation size once shells become densely populated. The implication is that small fractional reductions in average post-mission dwell time, applied across 10^4 -class fleets, dominate the cumulative collision-risk integral over multi-decade horizons. This motivates the fleet-scaling analysis of Section VIII-G.

III. SYSTEM ARCHITECTURE

A. Layered Decomposition

EcoOrbit decomposes the de-orbit problem into three loosely coupled layers, summarized in Table I.

TABLE I

LAYERED DECOMPOSITION OF THE ECOORBIT PIPELINE

Layer	Function	Implementation
Physics	Orbital decay propagation	MATLAB Euler integration
Intelligence	RUL regression and inference	Python / scikit-learn
Decision	Multi-objective de-orbit timing	Weighted cost minimization

Each layer is independently testable and exposes a typed interface to the layer above it.

B. Data Flow

The pipeline ingests a satellite state vector and platform parameters—initial altitude h_0 , initial orbital velocity v_0 , mass m , drag coefficient C_D , and reference cross-sectional area A —together with an atmospheric density profile $\rho(h)$. The MATLAB physics layer integrates the equations of motion under quadratic drag with one-hour time steps until the spacecraft crosses an effective re-entry altitude of 120 km. The resulting $[t, h, v, F_D]$ trajectory is exported as comma-separated values, ingested by Python via *pandas*, and used to fit a degree-three polynomial in altitude. The trained model is serialized for fleet inference. The decision layer queries the trained model for many candidate de-orbit days and selects the day that minimizes the weighted cost function defined in Section VII.

C. Reproducibility and Performance

All random number generators are seeded; all configuration parameters are version-controlled; and the entire pipeline executes end-to-end in under four seconds on a standard laptop. The trained surrogate returns RUL predictions in microseconds, enabling fleet-scale evaluation as discussed in Section VIII.

IV. MATHEMATICAL FOUNDATION

A. Atmospheric Drag Force

Below approximately 600 km altitude, the dominant non-gravitational perturbation acting on a satellite is aerodynamic drag. The drag force F_D acting on a body moving at speed v through a fluid of density ρ with reference area A and drag coefficient C_D is

$$F_D = \frac{1}{2} \cdot C_D \cdot A \cdot \rho(h) \cdot v^2 \quad (1)$$

Equation (1) is quadratic in velocity and linear in atmospheric density. Because v decreases monotonically as the orbit decays while ρ grows quasi-exponentially, the drag-force history exhibits an explosive growth in the final hours before re-entry, captured directly by the numerical propagator.

B. Exponential Atmospheric Density

EcoOrbit adopts the standard exponential atmosphere

$$\rho(h) = \rho_0 \cdot \exp[-(h - h_0) / H] \quad (2)$$

where $H \approx 50$ km is a representative scale height in the LEO regime, and $\rho_0(500 \text{ km}) \approx 6.97 \times 10^{-13} \text{ kg m}^{-3}$ [8]. The exponential model is selected for its analytical tractability; its limitations—absence of diurnal bulge, of solar-cycle modulation through F10.7, and of geomagnetic-storm response through A_p —are revisited in Section X.

C. King-Hele Analytical Lifetime

As an analytical reference, EcoOrbit uses King-Hele's closed-form expression for the lifetime of a near-circular orbit under exponential atmospheric drag [7]:

$$t_{\text{RUL}} = (m / (C_D A \rho_0)) \cdot \sqrt{(a_0 / \mu)} \cdot (1 / H) \cdot [1 - e^{-(\Delta h / H)}] \quad (3)$$

where a_0 is the initial semi-major axis, Δh is the altitude drop to be considered, and $\mu = 3.986 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$ is the Earth's gravitational parameter. Equation (3) provides a roughly 10 ms reference computation against which the numerical and AI estimators can be triangulated.

D. Equations of Motion

For a near-circular decaying orbit, the secular evolution of the semi-major axis under drag is well approximated by

$$da/dt = -\rho(h) \cdot v \cdot a / \beta \quad (4)$$

where $\beta = m / (C_D A)$ is the ballistic coefficient. Equation (4) is integrated together with the instantaneous circular-orbit relation $v = \sqrt{\mu / a}$ to obtain $h(t) = a(t) - R_E$, where $R_E = 6,378$ km. The decoupling of secular and short-period dynamics is consistent with the one-hour integration step adopted in Section V.

V. NUMERICAL IMPLEMENTATION (MATLAB)

A. Euler Integration Scheme

The MATLAB physics layer integrates equation (4) using a forward Euler scheme with fixed step size $\Delta t = 3,600$ s. At each step the propagator (i) evaluates atmospheric density via equation (2); (ii) computes drag force via equation (1); (iii) updates the semi-major axis via equation (4); and (iv) recomputes orbital velocity. The choice of Euler over higher-order schemes is deliberate: it preserves transparency, makes the dominant source of numerical error easy to characterize (Section VIII), and remains computationally negligible relative to the model errors introduced by the exponential atmosphere.

B. Initial Conditions and Constants

The reference scenario, summarized in Table II, follows a 250 kg LEO small satellite at 500 km circular altitude—representative of a wide class of Earth-observation and technology-demonstration platforms.

TABLE II
REFERENCE SPACECRAFT AND ENVIRONMENTAL PARAMETERS

Symbol	Value	Description
m	250 kg	Spacecraft mass
C_D	2.2	Drag coefficient
A	0.03 m^2	Reference cross-section
h_0	500 km	Initial altitude
ρ_0	$6.97 \times 10^{-13} \text{ kg/m}^3$	Density at 500 km
H	50 km	Scale height
μ	$3.986 \times 10^{14} \text{ m}^3/\text{s}^2$	Earth gravitational parameter
R_E	6,378 km	Earth equatorial radius
Δt	3,600 s	Integration step
h_{re}	120 km	Re-entry altitude

All constants are version-controlled in a single configuration file.

C. Termination Criteria and Logging

Integration terminates when $h(t) < 120$ km, defined here as the effective re-entry boundary. At each step the propagator logs altitude, velocity, drag force, and elapsed time; the resulting time series is exported as a CSV record consumed by the Python intelligence layer.

D. Computational Performance

On a representative consumer-class CPU, end-to-end propagation from $h_0 = 500$ km to re-entry executes in approximately four seconds. Although adequate for single-spacecraft analysis, this cost is prohibitive when applied per-satellite to a 10^4 -class constellation, motivating the AI surrogate of Section VI.

E. Pseudocode

Algorithm 1 summarizes the propagator. The state vector $\mathbf{x} = [a, v, h, F_D]$ is updated each step; integration terminates when h drops below the re-entry threshold.

Algorithm 1: Numerical orbital decay propagator

```

Input:  $h_0, m, C_D, A, \rho_0, H, \mu, R_E, \Delta t, h_{\text{re}}$ 
Output: trajectory  $T = [(t_i, h_i, v_i, F_{D_i})]$ 
 $a \leftarrow R_E + h_0$ ;  $v \leftarrow \sqrt{\mu / a}$ ;  $t \leftarrow 0$ ;  $T \leftarrow []$ 
while  $(a - R_E) > h_{\text{re}}$  do
   $h \leftarrow a - R_E$ 
   $\rho \leftarrow \rho_0 \cdot \exp(-(h - h_0)/H)$ 
   $F_D \leftarrow 0.5 \cdot C_D \cdot A \cdot \rho \cdot v^2$ 
   $da \leftarrow -\rho \cdot v \cdot a \cdot C_D \cdot A / m \cdot \Delta t$ 
   $a \leftarrow a + da$ 
   $v \leftarrow \sqrt{\mu / a}$ 
   $t \leftarrow t + \Delta t$ 
  append  $(t, h, v, F_D)$  to  $T$ 
end while
return  $T$ 

```

VI. ARTIFICIAL-INTELLIGENCE INTELLIGENCE LAYER

A. Training-Data Generation

The intelligence layer treats the MATLAB propagator as an oracle. A single, deterministic propagation produces a paired sequence (h_i, t_i) that constitutes the training set: each altitude observation is annotated with the simulator's prediction of the time required to fall from h_i to the re-entry boundary. Because the propagator is deterministic and seeded, the surrogate inherits exact reproducibility.

B. Polynomial Regression Model

EcoOrbit fits a degree-three polynomial in altitude to predict RUL:

$$RUL(h) = c_0 + c_1 h + c_2 h^2 + c_3 h^3 \quad (5)$$

Coefficients are estimated by ordinary least squares using *scikit-learn*'s *LinearRegression* class with degree-three polynomial features. The model is intentionally simple: a handful of coefficients per spacecraft class, no hidden state, and direct human inspectability of the response surface. This is consistent with the broader sustainability ethos of the project, which favors auditability over marginal accuracy gains.

C. Hyperparameter Selection

Polynomial degree was selected by leave-one-out cross-validation over candidate degrees $d \in \{1, 2, 3, 4, 5\}$. Degree three minimized RMSE while avoiding the high-frequency oscillations characteristic of higher-degree fits to monotone targets. Higher-degree models did not yield meaningful generalization improvement once the residual was dominated by atmospheric-model error.

D. Validation Metrics

Two scalar metrics summarize fit quality: the coefficient of determination R^2 and the root-mean-square error (RMSE) measured in days. On the reference trajectory, the trained surrogate achieved $R^2 = 0.9828$ and $\text{RMSE} = 8.58$ days. Inference latency, measured over 10^6 random altitudes, averaged $1.2 \mu\text{s}$ per query.

E. Pseudocode

Algorithm 2 summarizes the training and inference cycle of the surrogate. The training set is generated once per spacecraft class, and the resulting model is serialized for re-use across the operator's fleet.

Algorithm 2: RUL surrogate training and inference

```

Input: trajectory  $T = [(t_i, h_i)]$ 
Output: trained polynomial model  $\hat{M}$ 
 $X \leftarrow [h_i]; y \leftarrow [t_{\text{end}} - t_i]$   $\triangleright$  time-to-reentry targets
 $\Phi(X) \leftarrow$  polynomial features up to degree 3
 $\hat{M} \leftarrow \text{LinearRegression}().\text{fit}(\Phi(X), y)$ 
for  $d$  in  $\{1, \dots, 5\}$  do  $\triangleright$  leave-one-out CV
     $\text{RMSE}_d \leftarrow \text{cross\_val}(\hat{M}, d)$ 
end for
 $d^* \leftarrow \text{argmin}_d \text{RMSE}_d$  ( $\Rightarrow d^* = 3$ )
save  $\hat{M}$  to disk
inference:  $\text{RUL}(h) \leftarrow \hat{M}.\text{predict}(\Phi([h]))$ 

```

F. Auditability and Energy Cost

Polynomial regression admits exact closed-form coefficients and bounded inference cost. The serialized model occupies only a few kilobytes per spacecraft class, in contrast to the megabytes-to-gigabytes typical of deep neural surrogates [19]. Inference at 10^6 queries draws less than one watt-second on commodity CPUs, an important consideration for ground-segment systems serving large constellations.

VII. MULTI-OBJECTIVE DE-ORBIT OPTIMIZATION

A. Cost-Function Formulation

Let $f(t)$, $r(t)$, and $d(t)$ denote three normalized cost components: $f(t)$, the fuel/ ΔV budget required to initiate disposal at day t ; $r(t)$, the residual collision-probability integral over the time remaining in the operational orbit; and $d(t)$, the post-mission debris dwell time before atmospheric re-entry. Each component is normalized to the unit interval $[0, 1]$. EcoOrbit then minimizes the convex combination

$$J(t) = w_f f(t) + w_r r(t) + w_d d(t) \quad (6)$$

with $w_f = 0.4$, $w_r = 0.3$, and $w_d = 0.3$. The fuel-heavy weighting reflects the fact that operators control fuel directly, whereas risk and debris dwell are partly determined by the broader orbital environment.

B. Component Models

The fuel cost $f(t)$ is modeled as a monotonically decreasing function of t , since later disposal allows the orbit to decay naturally and reduces the required ΔV . The risk cost $r(t)$ is modeled as a monotonically increasing function, reflecting cumulative exposure to the conjunction environment over an extended residual life. The dwell cost $d(t)$ is computed directly from the AI surrogate of Section VI by integrating the post-mission decay until re-entry.

C. Search Procedure

Because each evaluation of $J(t)$ costs only a few microseconds via the surrogate, EcoOrbit performs an exhaustive sweep over candidate days $t \in \{1, 2, \dots, T_{\text{max}}\}$. The minimizer is $t^* = \text{argmin}_t J(t)$. For the reference scenario, $t^* = 200$ days, corresponding to a 62 percent reduction in residual debris dwell time relative to the unmanaged baseline.

D. Sensitivity to Weights

Sensitivity analysis perturbs each weight by ± 0.05 with the constraint $w_f + w_r + w_d = 1$. The optimal t^* shifts by less than ± 11 days across the perturbation envelope, indicating that the qualitative recommendation is robust to plausible operator preferences. Exposing the weights as user inputs allows mission planners to align disposal scheduling with their own risk-cost trade-offs.

E. Pseudocode

Algorithm 3: Multi-objective de-orbit optimization

```

Input: surrogate  $\hat{M}$ , weights  $(w_f, w_r, w_d)$ ,  $T_{\text{max}}$ 
Output:  $t^*$  (optimal de-orbit day)
 $J_{\text{min}} \leftarrow +\infty; t^* \leftarrow 0$ 
for  $t \in \{1, \dots, T_{\text{max}}\}$  do
     $f(t) \leftarrow f_{\text{model}}(t)$   $\triangleright \Delta V$  cost
     $r(t) \leftarrow r_{\text{model}}(t)$   $\triangleright$  collision risk
     $d(t) \leftarrow \hat{M}.\text{predict}(h(t))$   $\triangleright$  post-mission dwell
     $J(t) \leftarrow w_f \cdot f + w_r \cdot r + w_d \cdot d$ 
    if  $J(t) < J_{\text{min}}$  then
         $J_{\text{min}} \leftarrow J(t); t^* \leftarrow t$ 
    end if
end for
return  $t^*$ 

```

F. Comparison with State of the Art

Recent work has explored statistical-learning approaches to LEO lifetime prediction [19], typically using Gaussian-process regression or feed-forward networks. EcoOrbit differs in two respects. First, it integrates the surrogate inside an explicit, operator-tunable cost function rather than presenting it as a stand-alone predictor. Second, it preserves a closed-form analytical baseline (King-Hele) for cross-validation, which is

critical for regulatory acceptance. Table V summarizes the qualitative trade-off.

TABLE V
QUALITATIVE COMPARISON WITH PRIOR APPROACHES

Approach	Speed	Auditable
Numerical only [3], [8]	Slow	Yes
Analytical only [7]	Fast	Yes
Black-box ML [12], [19]	Fast	No
EcoOrbit (this work)	Fast	Yes

EcoOrbit recovers the speed of black-box surrogates without sacrificing the audibility that closed-form baselines provide.

VIII. RESULTS AND VALIDATION

A. Trajectory Reproduction

Figure 1 (described qualitatively) compares the MATLAB-propagated altitude history with the King-Hele closed-form lifetime. From an initial $h_0 = 500$ km, altitude decreases slowly during the first ~ 350 days, then declines rapidly as drag grows quasi-exponentially with falling altitude. Orbital velocity rises monotonically from 7.612 km s^{-1} to 7.846 km s^{-1} at re-entry, and drag force grows from approximately $6 \times 10^{-6} \text{ N}$ at injection to over $3 \times 10^{-3} \text{ N}$ in the final hours.

B. Triangulated RUL Estimates

Table III summarizes the three independent RUL estimators.

TABLE III
TRIANGULATED REMAINING-USEFUL-LIFE ESTIMATES

Method	RUL	Δ vs. Sim.
King-Hele analytical	580 days	+3.2%
Numerical (Euler, baseline)	562 days	—
AI polynomial regression	555 days	−1.3%

Three independent methods agree within 3.2 percent for the reference small-sat scenario.

The 3.2 percent inter-method spread is consistent with the expected error budget of the exponential-atmosphere assumption. The numerical propagator is taken as the operational baseline because it admits straightforward extension to higher-fidelity atmospheric models; the analytical and AI estimators serve as cross-checks.

C. Fit Quality of the AI Surrogate

The trained polynomial regression achieves $R^2 = 0.9828$ and $\text{RMSE} = 8.58$ days on the reference trajectory. Residuals are largely centered, with mild over-prediction of RUL near the injection altitude (where curvature in the response surface is small) and mild under-prediction near the terminal altitudes (where the exponential drag growth is most aggressive). These

residuals are below the noise floor introduced by atmospheric-model error and therefore do not justify additional model complexity at the current fidelity.

D. Optimal De-Orbit Day

Sweeping $J(t)$ over $t \in [0, 580]$ days produces a smooth, single-minimum cost curve. The minimum occurs at $t^* = 200$ days, at which $J(t^*) \approx 0.31$. Compared with a baseline strategy that delays disposal until natural re-entry, $t^* = 200$ days reduces residual debris dwell time by 62 percent, with only a modest increase in ΔV .

E. Trajectory Phases

Inspection of the propagated trajectory reveals three qualitatively distinct phases. In the long quiescent phase ($t \approx 0$ –350 days), drag deposits energy at a rate of order $10^{-5} \text{ W kg}^{-1}$; altitude decays approximately linearly at $\sim 0.6 \text{ km day}^{-1}$. In the transition phase ($t \approx 350$ –540 days), drag rises by an order of magnitude as altitude crosses ~ 350 km, and altitude decay accelerates to $\sim 1.5 \text{ km day}^{-1}$. In the terminal phase ($t > 540$ days), drag grows quasi-exponentially, altitude decay exceeds 5 km day^{-1} , and the spacecraft crosses the 120 km re-entry threshold within hours of crossing 200 km. The surrogate's largest residuals occur in the third phase, where the response surface has the steepest curvature; this is the dominant contributor to the reported 8.58-day RMSE.

F. Error Budget

Three sources contribute to the inter-method spread observed in Table III. Numerical truncation error from the Euler scheme accounts for approximately 1.5 percent of the lifetime estimate, dominated by the near-singular drag growth in the terminal phase. Atmospheric-model error, driven by the assumption of a single exponential scale height, contributes approximately 2.0 percent in the absence of solar-cycle modulation. Surrogate model error, in the form of polynomial-fit residuals over the [120 km, 500 km] altitude band, contributes a further 1.3 percent. The geometric sum of these error sources, ~ 2.8 percent, is consistent with the empirically observed 3.2 percent inter-method spread.

G. Inference Speed-Up and Fleet Scaling

Replacing the four-second numerical propagation with the trained surrogate yields more than a $10^6 \times$ reduction in inference time per spacecraft. As summarized in Table IV, the architecture scales naturally from a single satellite to a fleet of 10^4 .

TABLE IV
DEPLOYMENT TIERS

Tier	Scale	Cycle time
1: Single-satellite validation	1	~ 4 s
2: Operator fleet deployment	10^2	~ 0.1 s
3: Global LEO traffic manager	10^4	< 1 s

Compute scales linearly with constellation size; data ingestion becomes the practical bottleneck at Tier 3.

IX. DISCUSSION

The principal methodological contribution of EcoOrbit is the deliberate triangulation of an analytical baseline, a numerical propagator, and a learned surrogate. Each estimator contributes a distinct epistemic guarantee: King-Hele provides closed-form interpretability, the Euler integrator admits a controllable accuracy–cost trade-off, and the polynomial regressor provides scalable inference. Disagreement between estimators is itself an informative diagnostic; convergence within 3.2 percent on the reference scenario raises operational confidence in a way that no single estimator could provide on its own.

The decision layer's exposure of operator-tunable weights w_f , w_r , w_d turns disposal scheduling from an opaque heuristic into a structured negotiation. Different operators legitimately weigh fuel, risk, and dwell differently; allowing those preferences to be expressed numerically makes the resulting de-orbit recommendation defensible in front of regulators and insurers.

Finally, the deliberate choice of a low-capacity surrogate—polynomial rather than deep neural network—reflects a sustainability ethos that prizes auditability and energy-efficient inference. The marginal accuracy gains from a more expressive model are dominated, at the present level of fidelity, by the modeling error of the exponential atmosphere.

A. Operator Implications

From the operator's perspective, three properties of EcoOrbit are immediately consequential. First, the recommendation is deterministic and reproducible, supporting audit trails for insurance and regulatory filings. Second, the multi-objective formulation lets a fleet manager translate strategic policy—conservative versus aggressive disposal—directly into the weight vector (w_f, w_r, w_d) , eliminating the need for ad-hoc rule lookups. Third, the surrogate's microsecond inference latency means that each conjunction-warning event can trigger a fresh re-optimization, rather than relying on stale ground-segment heuristics.

B. Regulatory and Insurance Considerations

Recent regulatory developments, including the U.S. Federal Communications Commission's adoption of a five-year post-mission disposal rule for LEO satellites [20], are tightening the compliance environment. EcoOrbit complements such guidance by quantifying the cost of compliance: an operator can compute the marginal ΔV required to satisfy a five-year window, and offset it against expected reductions in insurance premia driven by lower residual collision exposure. This monetizes orbital sustainability in a way that aligns commercial and environmental incentives.

C. Threat to Validity

The principal threat to validity of the present results is the use of a single reference scenario. The 250 kg, 500 km circular case is representative but does not exercise the surrogate at the extremes of mass, drag area, or altitude. A multi-class corpus is required before the headline numbers (562-day RUL, 200-day t^* , 62 percent dwell saving) can be claimed at fleet level. That work is delineated in Section X.

D. Failure Modes and Defensive Engineering

Three plausible failure modes were considered. First, surrogate extrapolation outside the training altitude band would generate non-physical RUL predictions; the implementation rejects queries that lie outside [120, 600] km. Second, a stale F10.7 forecast would bias lifetime predictions during periods of rising solar activity; the architecture exposes a configurable refresh interval for the density model. Third, an operator could induce pathological behavior by setting two of the three weights to zero; $J(t)$ is therefore guarded against degenerate weight vectors and falls back to a conservative IADC-aligned default when (w_f, w_r, w_d) is ill-specified. Together these mitigations support a defensive engineering posture appropriate for a system that influences high-value disposal decisions.

X. LIMITATIONS AND FUTURE WORK

A. Numerical-Integration Error

The forward-Euler scheme accumulates approximately 3 percent error relative to a converged reference. Replacing Euler with a fixed-step fourth-order Runge-Kutta (RK4) integrator is expected to reduce numerical truncation error to below 0.1 percent at modest computational cost.

B. Atmospheric Model Fidelity

The exponential atmosphere ignores the diurnal density bulge, the eleven-year solar cycle modulated by the F10.7 index, and

short-term geomagnetic storms parameterized by the Ap index. Substituting NRLMSISE-00 with live F10.7 and Ap feeds [14] would significantly tighten lifetime predictions, particularly during periods of high solar activity.

C. Spacecraft Class Generalization

The current surrogate is trained on a single reference trajectory. Generalization to multi-class fleets—where m , C_D , and A vary by spacecraft—requires retraining on a corpus that spans the relevant ballistic-coefficient range. A multi-input regressor with parameters (h , m , C_D , A) is a natural next step.

D. Inter-Satellite Coordination

EcoOrbit currently optimizes per-satellite. A fleet-coordinating layer that schedules disposal across satellites to minimize cumulative collision probability—an instance of stochastic combinatorial optimization—would extract additional sustainability value at the operator level.

E. Live Data Integration

Production deployment requires ingestion of public conjunction data, including TLEs from CelesTrak and high-precision ephemerides from the U.S. Space Force's Space-Track service [15], together with covariance information used by conjunction-assessment services such as ESA's CREAM [16]. The current architecture exposes an ingestion port for these feeds.

F. Uncertainty Quantification

The present formulation reports point estimates of RUL and t^* . A natural next step is to wrap the cost function in a Monte-Carlo loop over uncertain inputs, particularly C_D (typically 2.0–2.4 with significant epistemic uncertainty for tumbling end-of-life satellites) and the F10.7 forecast (whose multi-year envelope spans more than a factor of two). Operators would then receive a distribution over t^* rather than a deterministic recommendation, supporting risk-tolerance-driven decision-making.

G. Coupling to Conjunction-Assessment Services

Long-term, EcoOrbit can be exposed as a sustainability-API consumed by commercial conjunction-assessment providers and federated with services such as Space-Track and CREAM. In this configuration, every conjunction warning generated against a managed satellite would also trigger a fresh re-evaluation of t^* , allowing the disposal recommendation to evolve with the operational risk picture rather than remaining static across the mission.

XI. SUSTAINABILITY IMPACT AND SDG ALIGNMENT

The contributions of EcoOrbit map directly to three United Nations Sustainable Development Goals. SDG 9 (Industry, Innovation, and Infrastructure) is advanced by modernizing satellite end-of-life management with predictive analytics, raising the maturity standard of orbital infrastructure stewardship. SDG 12 (Responsible Consumption and Production) is supported by extending the productive life of each spacecraft and by shortening post-mission debris dwell, thereby reducing the per-satellite environmental footprint of the space industry. SDG 13 (Climate Action) is supported indirectly: Earth-observation satellites underpin climate monitoring, and protecting the LEO regime from cascading collisions safeguards the very assets that monitor planetary change.

Quantitatively, the reference scenario removes 62 percent of post-mission debris dwell time per managed satellite. Generalized across an operator-class fleet of 10^2 spacecraft, this corresponds to thousands of satellite-years of avoided collision exposure; at the constellation scale of 10^4 satellites, the cumulative reduction in orbital debris dwell becomes globally significant.

Beyond the per-satellite metric, EcoOrbit also contributes to the broader sustainability discourse by demonstrating that auditable, energy-efficient AI surrogates are sufficient for operationally meaningful orbital decision-making. This stands in deliberate contrast to a literature trend toward ever-larger machine-learning models. The total energy required to train and deploy an EcoOrbit instance for one spacecraft class is dominated by the four-second numerical propagation; the polynomial fit, the model serialization, and a million-query inference run together consume less than one CPU-second. For a 10^4 -class constellation re-evaluated daily, the cumulative compute footprint remains under ten kilowatt-hours per year—a small fraction of the operational power budget of a single ground antenna.

This footprint analysis is consistent with the broader principle that sustainability technologies should not themselves impose disproportionate environmental costs. By preserving auditability, capping inference cost, and exposing decision weights, EcoOrbit aligns the technical design with the policy intent of the IADC and UNOOSA frameworks [6], [22].

XII. CONCLUSION

This paper has presented EcoOrbit, a hybrid physics–AI decision engine that converts end-of-life de-orbit timing for LEO small satellites from a reactive operation into a predictive, multi-objective optimization. A MATLAB Euler propagator integrates the equations of orbital decay under quadratic atmospheric drag; a Python degree-three polynomial regression accelerates inference by more than a factor of 10^6 while preserving $R^2 = 0.9828$ and 8.58-day RMSE; and a weighted cost function balances fuel, risk, and debris dwell to identify the optimal disposal day. Triangulation against the King-Hele analytical lifetime confirms inter-method agreement to 3.2 percent.

For the reference 250 kg LEO platform, EcoOrbit recommends de-orbit at day 200, removing 62 percent of post-mission debris dwell time. The architecture scales from a single spacecraft to fleets of 10^4 , supporting near-term constellation-wide deployment and long-term integration with global LEO traffic-management services. Looking ahead, the planned migration to RK4 integration, NRLMSISE-00 atmospheric modeling with live solar and geomagnetic indices, multi-class surrogate training, and Monte-Carlo uncertainty quantification will tighten lifetime predictions and supply operators with calibrated probability distributions over disposal dates rather than point estimates. Coupled with operator-tunable cost weights, this trajectory positions EcoOrbit as a credible building block for the predictive, sustainability-aware orbital traffic management that the next decade of LEO operations will demand.

APPENDIX

A. Derivation of the Secular Decay Rate

Starting from the energy of a circular orbit $E = -\mu m / (2a)$, and noting that the rate at which drag does work on the spacecraft is $dE/dt = -F_D \cdot v$, we equate

$$\mu m / (2 a^2) \cdot da/dt = -\frac{1}{2} C_D A \rho v^3 \quad (\text{A.1})$$

Substituting the circular-orbit relation $v^2 = \mu / a$ yields equation (4) of the main text. The same derivation, using $v \approx \sqrt{\mu/a_0}$ as a constant first approximation, yields the King-Hele closed-form lifetime in equation (3).

B. Density at Reference Altitude

The 500 km reference density $\rho_0 = 6.97 \times 10^{-13}$ kg m $^{-3}$ adopted in the present work is consistent with mid-solar-cycle values from the U.S. Standard Atmosphere [21] and with the long-term mean of NRLMSISE-00 [14] over altitudes 480–520 km.

Variability over the eleven-year solar cycle is approximately a factor of four, which is the principal source of long-horizon prediction uncertainty for orbital lifetime in LEO.

C. Ground-Segment Software Stack

The reference implementation uses MATLAB R2024a for the propagator, Python 3.11 with NumPy, pandas, scikit-learn, and SciPy.optimize for the AI and decision layers, and Filter Analyzer for spectral diagnostics. The complete pipeline is reproducible from a single configuration file; all outputs—trajectory CSVs, fitted model artifacts, and decision logs—are checksummed and version-controlled.

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